

Metric-KNN is All You Need

Tejas Anvekar

CEVI @ KLE Technological University
Vidyanagar, Hubballi, Karnataka, India
anvekar.tejas@gmail.com

Dikshit Hegde

CEVI @ KLE Technological University
Vidyanagar, Hubballi, Karnataka, India
dikshithegde@gmail.com

Ramesh Ashok Tabib

CEVI @ KLE Technological University
Vidyanagar, Hubballi, Karnataka, India
ramesh_t@kletech.ac.in

Uma Mudengudi

CEVI @ KLE Technological University
Vidyanagar, Hubballi, Karnataka, India
uma@kletech.ac.in

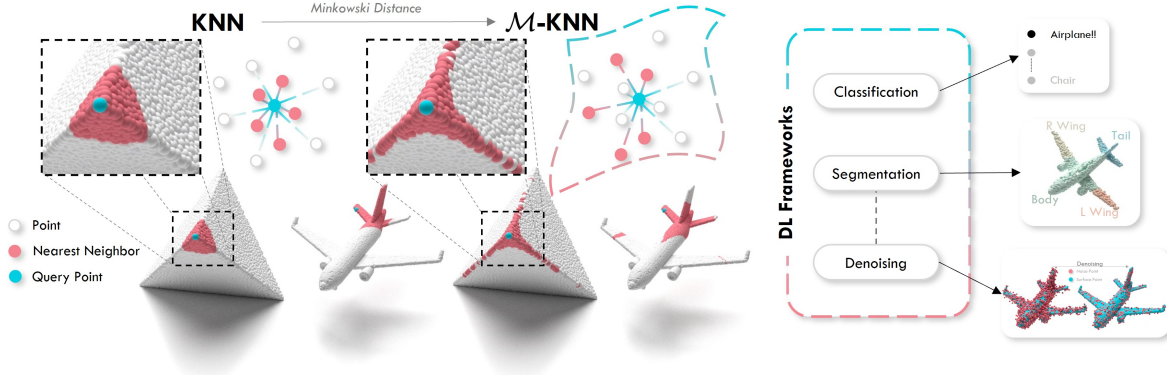


Figure 1: Metric-KNN (M -KNN) is a topology aware Nearest-Neighbor search algorithm. The figure shows topology aware learning facilitates Deep-Learning models to imitate human-perception. Highlighted regions of tetrahedron depicts nearest-neighbor selection similar to human-perception. M -KNN lays foundation for wide-range of 3D Deep-Learning models towards topology aware learning for downstream task like classification, part segmentation, and denoising of point clouds.

ABSTRACT

In this work, we propose a novel Metric-K Nearest Neighbor (M -KNN) to facilitate topology aware learning in point clouds. Topology aware learning is achieved by accumulation of local features in deep-learning model. Recent work rely on Ball queries or K-Nearest-Neighbor (KNN) for local feature extraction of point clouds and finds challenges in retaining topological information. M -KNN employs a generalised Minkowski distance in the KNN search algorithm for topological representation of point clouds. M -KNN enables state-of-the-art point cloud methods to perform topology aware downstream tasks. We demonstrate the performance of M -KNN as plugin towards point cloud classification, part-segmentation, and denoising using benchmark dataset.

KEYWORDS

Point cloud representation, Classification, Segmentation, Denoising, K Nearest Neighbour

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1 INTRODUCTION

In recent years, point clouds have gained significance due to their sparse and unordered data. Point cloud facilitates faster processing with minimal memory footprints for a wide range of real-world applications, including 3D object detection, AR/ VR/ Meta-verse, and self-driving automobiles. For deep-learning models to be interpretable and robust, 3D data must be processed effectively and analytically. The data as point cloud can be considered as both boon and curse for learning models, as it lacks topological/ geometric information. State-of-the-art models rely on Ball-Query or KNN for local information. Ball-Query/ KNN identifies local neighborhood in point clouds, unlike human-perception that identifies 3D world based on shape and structural primitives such as corners, edges, surfaces [Anvekar et al. 2022]. Towards enabling topology driven learning of point clouds, the main contributions are:

- We propose to imitate human-perception in point clouds representation, by introducing generalised Minkowski distance in KNN algorithm and name it as Metric-KNN (M -KNN).

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- We show the effect of $\mathcal{M}$ -KNN on popular point cloud classification, segmentation and denoising methods.
 - Towards point cloud classification and segmentation, we propose $\mathcal{M}$ -KNN version of DGCNN [Wang et al. 2019].
 - We propose $\mathcal{M}$ -KNN version of DMR [Luo and Hu 2020] towards point cloud denoising.
- We demonstrate the performance of proposed methodology on benchmark dataset (ShapeNet[Chang et al. 2015]) and compare the results with state-of-the-art techniques.

2 METRIC K-NEAREST NEIGHBOUR

We introduce, $\mathcal{M}$ -KNN for topology aware point cloud learnt representations. KNN or Ball-Query identifies local neighbour unlike human-perception that identifies shapes, structural geometry, corners, etc. as shown in highlighted region in Figure 1. Given xyz coordinates of N points $\mathbf{P}$: $\mathbf{p} \in \mathbb{R}^3$. K topological local neighbors in $\mathbf{P}$ can be given by equation:

$$\mathcal{M}\text{-KNN}(\mathbf{P}, K) = \mathbf{P} \left[\underset{k \in (1, \dots, K)}{\operatorname{argmin}} \|\mathbf{p}_i - \mathbf{p}_j\|_r \right] \quad (1)$$

where $(\mathbf{p}_i, \mathbf{p}_j \in \mathbf{P})$ and $(i, j \in N)$. $\mathcal{M}$ -KNN employs Minkowski distance of order r and is given by $\|\mathbf{p}_i - \mathbf{p}_j\|_r$. In Equation 1, substituting $r = 2$ yields KNN graph.

Towards topology aware learning of point clouds, we demonstrate $\mathcal{M}$ -KNN to be incorporated with the most widely used models. Introducing $\mathcal{M}$ -KNN overcomes the drawbacks of edge-convolutions, towards extraction of edge and corner information from point clouds. We demonstrate the effects of $\mathcal{M}$ -KNN on supervised and unsupervised models and emphasise on DGCNN and DMR respectively as both of them incorporate *EdgeConv*.

3 RESULTS AND DISCUSSIONS

In this section, we discuss the effect of $\mathcal{M}$ -KNN on dataset used and compare with state-of-the-art methods. To evaluate the performance of the proposed methodology we use benchmark dataset ShapeNet towards point cloud classification, part-segmentation with DGCNN, and denoising with DMR.

We show topology aware points that are gathered in proposed methodology on tetrahedron and table as shown in Figure 1 and Figure 2. For a given **query point**, KNN gathers **neighbourhood points** on euclidean metric, where as $\mathcal{M}$ -KNN gathers points on semantically-similar or topological metric. We demonstrate the effect of $\mathcal{M}$ -KNN by substituting $r = \frac{1}{32}$ in Equation 1 heuristically for classification, part-segmentation and denoising of point cloud. We demonstrate and discuss the in-depth effect of $\mathcal{M}$ -KNN for a given learning model in Figure 2. For classification as a downstream task, we achieve **1.1%** improvement in mean accuracy, and a **0.87%** increase in overall accuracy over DGCNN. For segmentation as downstream task, we achieve **0.32%** rise in mIoU within 150 epochs. For denoising as downstream task, we show our approach has **three folds** lesser validation loss compared to DMR on varying embedding noise (0.01, 0.03, and 0.08), and visualization of the same is shown in Figure 1. We infer from results that the topology aware points gathering plugin ($\mathcal{M}$ -KNN) outperforms the state-of-the-art methods (DGCNN and DMR).

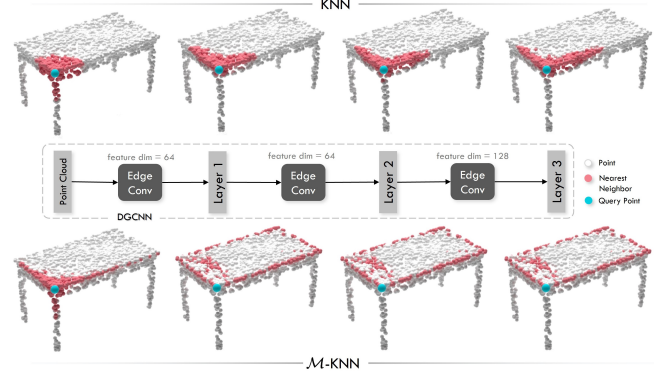


Figure 2: Layer-wise feature space visualization of DGCNN employed with KNN and $\mathcal{M}$ -KNN. Top row depicts KNN feature space and bottom row depicts $\mathcal{M}$ -KNN feature space (input and layers 1-3 are listed from left to right). We infer, $\mathcal{M}$ -KNN learns topological features in first layer (like table-edge, table-border, table-surface), where as KNN finds challenges even at layer 3. $\mathcal{M}$ -KNN version facilitates faster computation with reduced memory footprint via pruning of redundant feature-layers 2-3.

4 CONCLUSIONS

In this work, we have proposed a novel Metric-KNN ($\mathcal{M}$ -KNN) to facilitate topology aware learning in point clouds. $\mathcal{M}$ -KNN employs a generalised Minkowski distance in KNN search algorithm for semantically-similar or topological grouping in point clouds. $\mathcal{M}$ -KNN effectively compresses state-of-the-art methods; while preserving topological representations. Our approach facilitates deployment of efficient and compressed models on edge devices. We have demonstrated the performance of $\mathcal{M}$ -KNN as plugin towards point cloud classification, part-segmentation, and denoising using ShapeNet dataset.

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REFERENCES

- Tejas Anvekar, Ramesh Ashok Tabib, Dikshit Hegde, and Uma Mudengudi. 2022. VG-VAE: A Variational Geometry Point-Cloud Variational Auto-Encoder. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. 2978–2985.
- Angel X. Chang, Thomas Funkhouser, Leonidas Guibas, Pat Hanrahan, Qixing Huang, Zimo Li, Silvio Savarese, Manolis Savva, Shuran Song, Hao Su, Jianxiong Xiao, Li Yi, and Fisher Yu. 2015. *ShapeNet: An Information-Rich 3D Model Repository*. Technical Report arXiv:1512.03012 [cs.GR]. Stanford University – Princeton University – Toyota Technological Institute at Chicago.
- Shitong Luo and Wei Hu. 2020. Differentiable Manifold Reconstruction for Point Cloud Denoising. *CoRR* abs/2007.13551 (2020). arXiv:2007.13551 <https://arxiv.org/abs/2007.13551>
- Yue Wang, Yongbin Sun, Ziwei Liu, Sanjay E. Sarma, Michael M. Bronstein, and Justin M. Solomon. 2019. Dynamic Graph CNN for Learning on Point Clouds. *ACM Transactions on Graphics (TOG)* (2019).